

Domain Environment

IT Infrastructure Management (W04IST-SI0826G)

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This week we discuss:

- Domain environment
- Active Directory
- Managing organizational units and user accounts

Domains and Workgroups

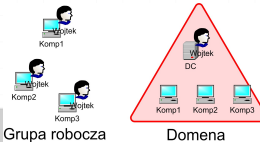
Workgroup

- Each computer configured independently
- Separate user account on every computer used

Domain

- Computers configured centrally
- Configuration stored in the domain
- One user account in the domain

Introduction



Workgroup

- Client (personal computer) (Windows 11)
- Server (e.g., Windows Server 2025)

Domain

- Client / personal computer
- Server

- LDAP
- Kerberos
- DFSR
- and many others

Kerberos Service

The domain controller provides the **Kerberos** service that controls single sign-on and authentication.

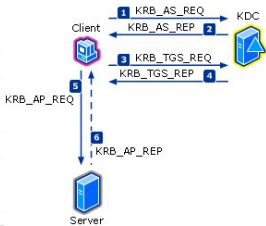
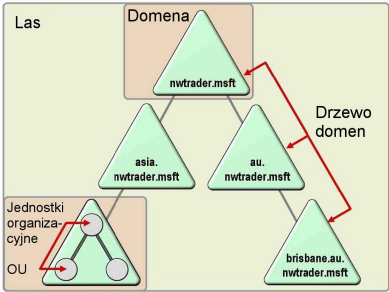


Figure 2: Kerberos Service supports single sign-on

Multi-Domain Environment



Physical elements

- Domain controller
- Data store
- Global catalog
- RODC

Logical elements

- Partitions
- Schema
- Domains
- Trees
- Forests
- Sites

Active Directory as a Directory Service

- Storing domain configuration
- Naming and describing resources
- Discovering resources
- Managing resources
- Securing resources

Advantages of Active Directory

- Integration with DNS
- Scalability
- Centralized management
- Decentralized management (delegation of permissions)

- 1 Install the **Active Directory Domain Services** role
- 2 Promote the server to the **domain controller** role

Installation using the PowerShell:

```
1 Install-WindowsFeature AD-Domain-Services
2 Import-Module ADDSDeployment
3 # For the first domain controller in the forest
4 Install-ADDSForest -DomainName contoso.internal -InstallDNS
5 # For an additional domain controller in the existing domain
```

Installation from Removable Media

On a domain controller run the command:

```
1 ntdsutil
2 activate instance ntds
3 ifm
4 create SYSVOL full C:\IFM
```

Move the generated file to the new computer and use it in the AD configuration wizard.

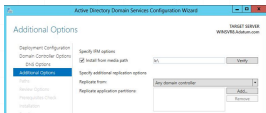


Figure 4: Importing configuration from the file

Managing Organizational Units

- Graphical tools
- PowerShell (+ VS Code w/ PowerShell extension)
- CMD (Command Prompt)

- "Containers" for objects
- Delegation of administrative tasks

AD Object Names

■ LDAP Distinguished Name

```
1 OU=Research,DC=contoso,DC=internal
```

identifies any object in the domain

■ LDAP Relative Distinguished Name

```
1 OU=Research
```

identifies an object in the parent container or context

■ Canonical Name

```
1 contoso.internal/Research
```

Components of a Distinguished Name

- 1 **DC — domain component** — components of the domain name
- 2 **OU — organizational unit** — names of organizational units
- 3 **CN — common name** — other objects: groups, users, computers

Example for user Sarah:

```
1 "cn=Sarah,ou=Marketing,ou=Sweden,DC=contoso,DC=internal"
```

1 Organizational unit for the Sweden

```
1 dsadd ou "OU=Sweden,DC=contoso,DC=internal"
```

2 Organizational unit for the HR and payroll department in Sweden

```
1 dsadd ou "OU=HR and payroll,OU=Sweden,DC=contoso,DC=internal"
```

Spaces may appear in the OU name, therefore quotation marks help define where the

PowerShell (recommended)

```
1 New-ADOrganizationalUnit -Name Sweden -Path "DC=contoso,DC=internal"
2 New-ADOrganizationalUnit -Name Accounting -Path
   ↳ "OU=Sweden,DC=contoso,DC=internal"
3 New-ADOrganizationalUnit -Name "United Kingdom" -Path "DC=contoso,DC=internal"
4 New-ADOrganizationalUnit -Name "Accounting" -Path "OU=United
   ↳ Kingdom,DC=contoso,DC=internal"
5 Remove-ADOrganizationalUnit "OU=Sweden,DC=contoso,DC=internal"
```

Organizational Units vs Containers

Containers (containers for built-in groups)

- Users
- Computers
- Built-in

Differences between OU and container

- Containers cannot be nested
- Permissions cannot be assigned to containers
- Group Policy Objects (GPO) cannot be linked to a container

User Accounts

■ Active Directory acts as a directory (telephone book)

■ Account properties:

- Name
- General / Display name
- Account

- User Principal Name (UPN) or User logon name example: wojtek@contoso.com
- SAMid or User logon name (pre-Windows 2000) example: contoso\wojtek
- Account options: Password never expires, User must change password, User cannot change password, etc.

- Active Directory Users and Computers console
- Active Directory Administrative Center console
- PowerShell

```
1 New-ADUser Anna -SamAccountName Anna -UserPrincipalName "annas@contoso.com"  
   -Path "OU=Sweden,DC=contoso,DC=internal" -DisplayName "Anna Svensson"  
   -AccountPassword (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force)
```

- See you next week

Bibliography

- [1] A. Warren, *Exam ref 70-742 identity with windows server 2016*. Microsoft Press, 2017.

